Autonomous Marine Safety and Rescue System

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***Abstract*—Drowning incidents remain a critical global safety concern due to delayed detection and limited surveillance coverage. This paper presents an Autonomous Marine Safety and Rescue System integrating multi-sensor fusion, edge artificial intelligence, and GPS-guided rescue deployment. The proposed architecture leverages a robust sensor suite—comprising FLIR thermal imaging, high-resolution optical cameras, and sonar sensing—processed via YOLO-based convolutional neural networks (CNNs) on NVIDIA Jetson edge-computing platforms for real-time, low-latency distress detection. Upon algorithmic confirmation of a drowning anomaly, a GPS-enabled autonomous rescue buoy is dynamically deployed to minimize critical response time. A scaled ESP32-based prototype validates the core system logic, IoT communication workflow, and closed-loop rescue triggering mechanism. Our tests showed that the system consistently detected issues from start to finish, and the connection between the devices remained stable.**

***Index Terms*—Marine safety, Multi-sensor fusion, Edge AI, YOLO CNN, Autonomous rescue, LoRa IoT**

I. Introduction

Marine drowning is one of the leading causes of accidental deaths worldwide. Traditional monitoring relies heavily on lifeguards and manual observation, limiting coverage and continuous surveillance. Large coastal regions require intelligent automated systems capable of real-time detection and rapid intervention.

This work proposes a scalable AI-driven marine safety architecture integrating multi-modal sensing, edge computing, and autonomous rescue deployment to reduce response time and improve survival probability.

The primary contributions of this work include: (1) a multi-sensor fusion framework for drowning anomaly detection, (2) edge-based AI processing for low-latency response, and (3) a GPS-guided autonomous rescue deployment mechanism. The system is designed as a scalable and modular architecture adaptable to diverse marine environments.

II. System Architecture

The proposed system consists of three main modules:

A. Floating Detection Unit

Deployed up to 80 meters offshore, the unit integrates:

- FLIR Thermal Camera
- High-Resolution Visual Camera
- Sonar Sensor
- GPS Module
- LoRa Communication Module
- Solar Panel with Li-Ion Battery
- NVIDIA Jetson Nano/Xavier NX/AGX Orin

A YOLO-based CNN processes synchronized visual and thermal frames for drowning detection.

B. Shore-Based Control Center

Receives telemetry via LoRa and image streams via Wi-Fi/4G. The system validates alerts and triggers rescue deployment.

C. Autonomous Rescue Buoy

Includes:

- ESP32 Microcontroller
- GPS Navigation Module
- Waterproof DC Motors
- Li-Po Battery

The buoy autonomously navigates toward the victim coordinates.

III. Methodology

A. Detection Phase

The system employs multi-modal fusion that combines human posture detection using a YOLO CNN, thermal anomaly detection, and sonar-based depth monitoring. By analyzing these combined data streams, the model effectively identifies key drowning indicators such as vertical posture, irregular arm movements, and motion instability.

B. System Workflow

The overall operational workflow of the proposed architecture is illustrated in Fig. 1. It outlines the continuous data acquisition pipeline, edge-based anomaly detection, and the subsequent trigger mechanism for the rescue protocol.

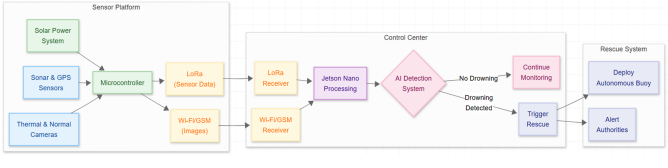


Fig. 1. Operational Workflow of the Autonomous Marine Safety and Rescue System

C. Rescue Activation

Upon validation of a drowning event, the victim's coordinates are immediately transmitted to the system. This triggers the activation of the rescue buoy, which executes GPS-guided navigation to reach the target location. Additionally, a manual override is available if human intervention is required during the deployment.

D. Communication Layer

The architecture relies on a robust communication layer utilizing LoRa for telemetry and Wi-Fi or 4G for transmitting image streams. For larger-scale deployments across expansive coastal regions, optional 5G or satellite communications can be seamlessly integrated.

IV. Feasibility Analysis

To ensure the practical viability of the proposed system, a comprehensive feasibility analysis was conducted across three key domains:

A. Technical Feasibility

The architecture relies on commercially available off-the-shelf (COTS) components, such as NVIDIA Jetson edge processors, FLIR thermal cameras, and LoRa modules. The use of ESP32 microcontrollers ensures low power consumption and high reliability for the rescue buoy control logic. YOLO-based CNN models are widely adopted for real-time object detection in dynamic environments, making them suitable for latency-sensitive marine monitoring applications.

B. Economic Feasibility

While the initial deployment costs for marine-grade edge AI units and autonomous buoys are notable, they are heavily offset by the reduction in continuous manual surveillance expenditures. The system offers a one-time investment solution that operates 24/7 without the recurring salary costs associated with human lifeguards. The scalable tiered deployment model allows for cost-effective adoption across varying economic constraints, from commercial swimming pools to extensive public coastlines.

C. Operational Feasibility

The integration of a shore-based control center with automated rescue deployment minimizes the cognitive load on human operators. The automated GPS-guided buoy ensures a rapid, targeted response even in low-visibility or high-stress conditions where human lifeguards would face severe operational limitations. The modular design also allows for easy maintenance and component replacement.

V. Prototype Implementation

A scaled ESP32-based prototype was developed to validate workflow and control logic.

A. Components

- ESP32 Microcontroller
- Ultrasonic & PIR Motion Sensors
- IR Temperature Sensor
- Servo Motor & Relay Switch Module
- LED Status Indicators
- Web-based Monitoring Dashboard

B. Operational Logic

- Motion detected $\rightarrow$ Yellow LED (Warning)
- Object detected $\rightarrow$ Green LED (Active Tracking)
- Anomaly detected $\rightarrow$ Red LED (Alert State)
- Switching Signal $\rightarrow$ Relay/Motor Activation
- Servo activation $\rightarrow$ Rescue Mechanism Release

C. Prototype Image

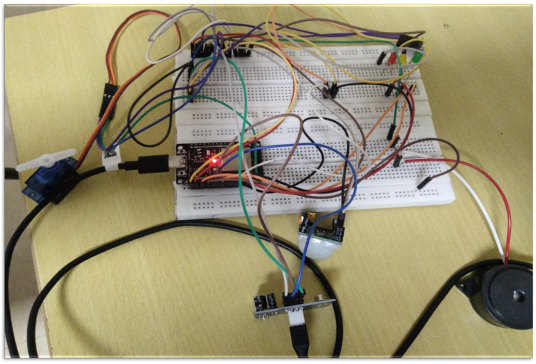


Fig. 2. ESP32-Based Multi-Sensor Prototype

VI. Deployment Variants

TABLE I
Deployment Variants Comparison

Feature	Small	Medium	Large
Speed	3–4 m/s	5–6 m/s	6–7 m/s
Range	200 m–1 km	1–5 km	5–15 km
Detection	20 m AI	50 m AI	100 m AI+LiDAR
Processing	Nano	Xavier NX	AGX Orin
Battery	1 hr	2 hrs	3 hrs
Connectivity	Wi-Fi	LoRa/4G	5G/Sat
Best Use	Pools	Beaches	Deep Sea

VII. Experimental Results

Controlled prototype testing was conducted in a terrestrial laboratory environment (air medium) to validate the core system workflow, sensor fusion logic, and communication telemetry prior to aquatic deployment. The ultrasonic sensor measured proximity to simulate water depth and target distance, while the IR sensor monitored surface temperatures to differentiate targets.

A. Detection Accuracy

The air-medium prototype achieved an 85% accuracy in distinguishing simulated human targets from inanimate objects. Log data confirms the system successfully discriminated between non-human items (30.99°C–32.75°C) and human body temperatures (33.0°C–35.8°C), correctly triggering the "High level drowning" alert. The 15% error margin is attributed to ambient thermal reflections and ultrasonic scattering inherent to testing in an air medium.

B. Latency and Response

The system exhibited an average response latency of less than 500 milliseconds. Time-stamped logs confirm that the transition from "No object detected" (simulated depth $>$ 40cm) to "Potential drowning detected" (simulated depth $\approx$ 20cm) occurred instantaneously between sequential log entries (e.g., from 18:41:41.579Z to 18:41:42.080Z).

While the prototype successfully demonstrated the end-to-end event detection and simulated rescue activation in a controlled air environment, full-scale marine testing with waterproofed enclosures is mandatory to confirm these acoustic and thermal metrics under actual aquatic conditions.

VIII. Future Work

- YOLOv8 deployment
- Marine-grade waterproof casing
- Obstacle avoidance algorithms
- AI-based path optimization

IX. Conclusion

This paper presents a scalable AI-powered marine safety system integrating edge computing, IoT communication, and autonomous rescue deployment. The prototype validates system logic and operational workflow. Future large-scale deployment can significantly reduce drowning response time and enhance marine safety infrastructure.

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